A Generative-AI Operating System for a Maglev ("Linear Shinkansen") in the Bada Quantum Language

Electromagnetic Levitation, a General-Relativity Manifold Layer, and the Jones Polynomial of a Thermal-Energy Body

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> ⚠️ **Scope / Honesty Note.** This is a **conceptual / simulation** report in the house style of the `Bada` repository. The electromagnetic levitation, the coil power budget, and the aerodynamic drag are ordinary, physically grounded models with correct SI numbers. The "anti-gravity field" is only the balancing magnetic field that holds the car up — a maglev does **not** switch gravity off. The "gravity-resistance cancellation via a general-relativity manifold integral" and the "Jones polynomial of the thermal-energy body" are the Yamaguchi TupleSpace motif used as control-gain and telemetry constructs. **Nothing here is a claim of new physics.**

要旨 / Abstract

本稿は、リニア新幹線のオペレーティングシステムとしての生成AIを、量子コンピューターの Bada 言語で記述した実装 `omega_maglev_os_pkg` を報告する。

We present a small generative-AI kernel, written in the conceptual **Bada** operator language and lowered to C, that acts as the operating system of a magnetically levitated train. Each control tick the kernel (i) commands an electromagnetic (EMS) levitation field so the lift force equals the car weight, (ii) budgets the coil Ohmic power as a normal-conductor **alternative to superconductivity**, (iii) models an evacuated-tube "no air resistance" running mode, (iv) trims a residual "gravity resistance" through the global partial-integral manifold element $1/(x \log x)^2$, and (v) tracks the Jones polynomial of a thermal-energy body via a Kauffman-bracket state sum weighted by a gamma-function factor. A π-softmax policy selects the control action and writes telemetry to an Akashic TupleSpace. We report the measured operating point (flux $B = 2.106$ T, amp-turns $NI = 16{,}761$, lift $= 441{,}299$ N matching the weight of a 45 t car) and state precisely which components are physical and which are conceptual.

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1. Introduction

Magnetically levitated ground transport replaces wheel-rail contact with a servo-controlled magnetic force. We ask a deliberately playful question in the project's idiom: *what would an "operating system" for such a vehicle look like if it were written as a generative controller* *in the Bada quantum language?* The answer is a five-subsystem kernel whose physical core (levitation, power, drag) is real and whose speculative shell (a manifold ”gravity resistance” term, a knot-theoretic thermal invariant) is the repository’s conceptual motif. We keep the two layers explicitly separated (§9).

2. The Bada operator algebra

Operator	Meaning
`<-`	left non-commutative act $\pi(\chi, x) = [i\pi,\ f(x)]$
`-<`	manifold integral $\iint 1/(x\log x)^2\ dx_m$
`>-`	quantum right act $e^{-x\log x}$
`Ω::push`	write to the Akashic TupleSpace `Ω::DATABASE`

The kernel `bada/maglev_os.bada` instantiates one class per subsystem and runs a single control tick `BadaOS <- shinkansen`.

3. Electromagnetic levitation (real physics)

An EMS pole develops an attractive force across the air gap g. With NI amp-turns,

$$B = \frac{\mu_0 NI}{g}, \qquad F = \frac{B^2 A}{2\mu_0} = \frac{\mu_0 (NI)^2 A}{2g^2},$$

where A is the pole area and $\mu_0 = 1.2566 \times 10^{-6}$ H/m. The controller solves $F = mg_0$ for the commanded amp-turns,

$$NI = g\,\frac{\sqrt{2mg_0}}{\mu_0 A}.$$

This is the honest core of what the marketing layer calls the ”anti-gravity field”: the field simply balances weight so the car floats.

4. A superconductor alternative

Rather than a superconducting coil ($R \to 0$), the kernel reports the Ohmic power a *normal* winding of T turns would dissipate,

$$P = I^2 R = \left(\frac{NI}{T}\right)^2 R,$$

so an operating point can be chosen that keeps P within the on-board power budget. For the reference car ($NI = 16{,}761$, $T = 400$, $R = 0.02\,\Omega$) we find $P \approx 35$ W per module.

5. Aerodynamics and the "no air resistance" mode

Drag is the textbook $F_{\mathrm{drag}} = \frac{1}{2}\rho C_d A_f v^2$. The only physical route to zero drag is to remove the air: an evacuated tube drops $\rho \to 0$. We report the removed fraction $r = 1 - \rho/\rho_{\mathrm{air}}$; at 500 km/h in vacuum, $F_{\mathrm{drag}} = 0$ and $r = 100\%$ versus sea level (42,541 N).

6. The general-relativity manifold layer (conceptual)

In real general relativity a free body follows a geodesic and there is no "gravity resistance" for a field to cancel. As a control-gain construct we integrate the global partial-integral manifold element

$$d\mu(x) = \frac{1}{(x\log x)^2}, \qquad M = \iint \frac{1}{(x\log x)^2}\, dx_m \approx \sum_{i=2}^{n+1} \frac{1}{(i\log i)^2},$$

and define a dimensionless cancellation ratio $c \in [0,1)$; the residual vertical acceleration reported to the OS is $a_{\mathrm{res}} = g_0(1-c)$. For 64 steps $M \approx 0.692$. *This term is a knob, not new physics.*

7. Jones polynomial of the thermal-energy body

The recirculating coolant / eddy-heat loop of a levitation module is represented as a knot diagram D. The Kauffman bracket is the state sum

$$\langle D \rangle = \sum_{s \in \{A,B\}^n} A^{a(s)-b(s)}\, d^{|s|-1}, \qquad d = -A^2 - A^{-2},$$

over all smoothings s of the n crossings, with $|s|$ the loop count. We fold in a gamma-function global-partial (beta) weight depending on body temperature T,

$$\beta(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \qquad J_{\mathrm{th}} = \langle D \rangle\, \beta\left(1 + \tfrac{T}{300},\ 1 + \tfrac{300}{T}\right).$$

For the trefoil model at $A = 1.2$, $T = 320$ K we obtain $\langle D \rangle = -6.9737$ and $J_{\mathrm{th}} = -1.1623$.

8. The generative-AI OS kernel

The kernel scores five actions {raise, lower field; evacuate tube; trim manifold; hold} from the physics error signals, normalises them, and forms a policy with the π-softmax

$$p_i = \frac{\exp(\ell_i\, \hbar_{\mathrm{eff}}\, \pi)}{\sum_j \exp(\ell_j\, \hbar_{\mathrm{eff}}\, \pi)}, \qquad \hbar_{\mathrm{eff}} = 0.35,$$

selecting $\operatorname{argmax}_i p_i$. Applying the action updates the gap, ambient density, or cancellation ratio, and the tick's telemetry is pushed to `Ω::DATABASE`.

9. Implementation and results

The package builds to a single `bin/launcher` (C99, `-lm`). Values below are direct program output.

Quantity	Value
Car mass	45,000 kg
Commanded amp-turns NI	16,761
Flux density B	2.1063 T
Lift force F	441,299 N ($= mg_0$)
Coil Ohmic power	35.1 W / module
Drag @ 500 km/h, vacuum	0 N (100 % removed)
Manifold integral M (64 steps)	0.6920
Kauffman $\langle D \rangle$ @ A=1.2	-6.9737
Thermal invariant J_{th} @ 320 K	-1.1623

The boot loop drives the residual-gravity trim first (largest error) and evacuates the tube once drag dominates, then holds a balanced state.

10. Scope and limitations

Physical and correct: the levitation equations (§3), the power budget (§4), and the drag model (§5). **Conceptual / metaphorical:** the manifold "gravity resistance" term (§6) and the mapping of heat flow to a knot (§7). The π-softmax kernel is a *simulated autopilot*, not a validated safety system and not a trained large model. No hardware claim is made.

11. Conclusion

We packaged a maglev "operating system" as a small generative controller in the Bada language, combining a correct electromagnetic-levitation core with the project's manifold and knot-theoretic motifs, and reported a self-consistent operating point. The value of the exercise is expository: it shows how far a physically honest core can be wrapped in the Bada operator idiom while keeping the speculative layer clearly labelled.

References

1. L. H. Kauffman, *State models and the Jones polynomial*, Topology **26** (1987) 395-407.
2. V. F. R. Jones, *A polynomial invariant for knots via von Neumann algebras*, Bull. AMS **12** (1985) 103-111.
3. M. Yamaguchi, *Bada Language, BadaOS and the TupleSpace framework*, Global Differential Manifold Research, 2025.